Air Quality (Carl Van Brink, Senior Associate Air Quality Consultant)

WSP will develop an Air Quality Impact Assessment (AQIA) report for the S7 Data Centre that is suitable for addressing the industry-specific SEARs for data centres, an inclusion in the SSD application to be submitted by Urbis.

This AQIA will include:

- An assessment of air quality impacts prepared in accordance with the NSW Environment Protection Authority (EPA) guidelines.
- An assessment of the potential for offsite dust impacts during construction works through completion of a Construction Dust Risk Assessment.
- An assessment of operational emissions conducted through plume dispersion modelling, this will focus on scenarios including:
  - Testing of backup power systems where emissions may occur (generators)
  - A peak emission incident (worst-case)
  - A likely emission incident
  - High-level mitigation options will be presented in the report
  - High-level monitoring and management measures will be presented in the report, depending on the modelling outcomes

Noise Impact Assessment (Camilo Chalela, Associate, VIC Acoustics Manager)

The Site Wide Noise and Vibration Impact Assessment report will include a site survey to determine the current noise environment on site and provide a basis for establishing operational and construction based on noise criteria applicable in NSW. Predictions of construction and operational noise levels at the nearest sensitive receivers will be carried out and compared against the established noise criteria. Anticipated vibration impacts during construction will be determined. Feasible mitigation measures will be developed to achieve compliance with criteria.

- Undertake a site survey to determine the current noise environment on site and provide a basis for establishing operational and construction noise criteria. This will include:
  - Unattended noise logging at one (1) representative location for a nominal period of 7 days. We require 7 fine weather days (i.e. no rain and <5m/s wind conditions) to undertake unattended noise monitoring and monitoring is to be undertaken outside of school holidays to represent typical noise levels.
  - Short term attended noise measurements will be conducted during the deployment of the logger to assist with the identification of major noise sources and the processing of unattended monitoring data.
- Establish the noise emission requirements in accordance with the relevant legislation, guidelines and standards (The Noise Policy for Industry 2017, Blacktown Development Control Plan 2015, Road Noise Policy 2011, NSW EPA Interim Construction Noise Guideline 2009, etc.)
- Review all project documentation and drawings including any manufacturer data relating to proposed equipment to be installed on site.
- Collate and review available acoustic data (ideally provided as 1/3 octave band sound power levels) for all acoustically significant plant and equipment (to be provided by the manufacturer/supplier).
- Prepare a 3D SoundPLAN® model of the site representative of the operational facility including:
  - Relevant noise emission/sound power data for all acoustically significant plant.
  - Available elevation data extending from the site to the closest noise sensitive receivers
  - Assess compliance with applicable operational noise limits at noise sensitive receivers.

— Where the predicted noise levels exceed the relevant criteria, suitable noise control measures will be recommended for discussion. Following approval, agreed measures will be modelled and assessed against nominated noise criteria.

— Review mechanical and electrical concept designs to identify any noise control considerations and space allowances that should be made for noise and vibration mitigation measures.

— Provide maximum sound power levels for generators, new chillers and other plant to meet environmental noise criteria at the nearest noise sensitive receivers.

— Carry out a preliminary construction noise and vibration impact assessment based on assumed construction scenarios, staging and equipment.

— Prepare a Noise Impact Assessment Report for SSDA submission

Environmental Sustainability Design (Matthew Payne – Director Sustainability and Climate Change)

— We will propose appropriate sustainability targets for agreement with the client and the design team. These may include:

  - Sustainability certifications
  - Operational targets – energy (PUE), water (WUE)
  - Embodied/upfront carbon targets
  - Climate risk strategy

— We will discuss and agree with the design team potential feasible design approaches that would support the achievement of the sustainability targets nominated to back check their achievability and appropriateness in the context of project constraints

— We will document the above in a report that responds specifically to the itemised requirements of the SEARS

— If required by the SEARS, preliminary quantified estimates may be developed, but can be typically achieved using desktop calculations or templates

— The Quantity Surveyor will be required to provide the data necessary for the completion of the NABERS Embodied Emissions Form/Template

— We do not typically need to attend site or collect any site data

Water Supply Infrastructure & Demand (Overview) (Jose Bergantinos, Senior Hydraulic Engineer)

— Existing Water Infrastructure and connection

- A DN200 mPVC is located north of the site, and;
- Connected to a DN375 DICL located northeast of the site along Honeycomb drive.
- Both mains are reticulating along Honeycomb drive
- Two connections is proposed frontage of the site.

— Potable Water Demand

- The estimated water demand for stage 1 is the following:
  - Domestic water usage: TBC
  - Industrial Water Use: TBC
  - Fire Hydrant: TBC
  - Fire Sprinklers: TBC
- Water Storage Tanks will be provided to the following:
  - Fire Tanks (Combined Sprinkler/Hydrant) – Capacity: TBC

  - Industrial Water Tanks – Capacity: TBC

— Water Infrastructure Capacity and Upgrades

- Subject to Water Services Coordinator – Section 73 (note: Hydraulic out of scope)

Sewer Infrastructure & Demand (Overview) (Jose Bergantinos, Senior Hydraulic Engineer)

— Existing Sewer Infrastructure

- A sewer manhole is located at the end of Honeycomb drive and Archbold Road with DN225 uPVC outlet.
- Sewer invert to finished surface level is 3.27mts.
- It is proposed that the sewer outflow of the site is connected to the existing sewer manhole.

— Sewer Demand

- The estimated sewer demand for stage 1 is the following:
  - Sewer outflow: TBC
  - Tradewaste: TBC

— Sewer Infrastructure Capacity and Upgrades

- Subject to Water Services Coordinator – Section 73 (note: Hydraulic out of scope)

Back-up Power System (Evan Pedlar, Technical Principal Electrical Engineer)

— Provide a detailed overview of any proposed back-up power system, including the scale and capacity of the system, and any associated testing procedures (frequency and duration).

- The overall project will include circa 320-340 x 3.1MW prime rated diesel generator sets to support the data hall IT and mechanical loads as well as, possibly, some smaller ancillary generator sets to support site-wide services.
- It is anticipated that the generators will be run-tested regularly and load tested annually via load bank. Load tests are typically up to 1h.

Due to the nature of the installation, it is expected that generators will be load tested one at a time and not simultaneously.

— Provide a detailed justification for the proposed back-up power system, including alternatives considered.

- It is not practicable to support loads of this magnitude or achieve the level of resilience or autonomy in backup power supply with alternative methods. This is based on the following:
- Insufficient space for renewable options due to rooftop being used entirely for mechanical plant and site building coverage being maximised.
- Backup autonomy (>48h) not practicable using alternative solutions.